

Operations Research Syllabus

Mohammed Brahimi



Instructors information

- Dr. Mohammed Brahimi
- Assistant Professor of Computer Science



- **Link to profile:**
 - <https://ensia.edu.dz/biography/mohammed.brahimi/>
- **Research interest:**
 - Machine learning and deep learning applied to real problems

- Dr. Soumaya Lakehal
- Assistant Professor of Mathematics



- **Link to profile:**
 - <https://ensia.edu.dz/biography/soumaya.lakehal/>
- **Research interest:**
 - Algorithms, Parallel Computing, Manufacturing Engineering, Heuristics, Logistics

Course description

*“ Operations Research (OR) module **connects theory and practice**, providing students with the **tools to model real-world problems** using linear programming and graph theory, and to **effectively solve these models and interpret their results**”*

Learning objectives

- Understand OR methodology for real-world problem-solving
- Understand Linear Programming (LP) and be able to use of solvers
- Familiarize the student with graph theory concepts
- Learn how to model a problem and solve it using LP and graph theory

Prerequisite

- Familiarity with basic mathematical concepts and techniques, such as algebra and calculus
- Knowledge of linear algebra, especially with regards to solving systems of linear equations

Course content (Part 1:Linear Programing)

1 The art of solving real-world problems: Operations Research (OR)

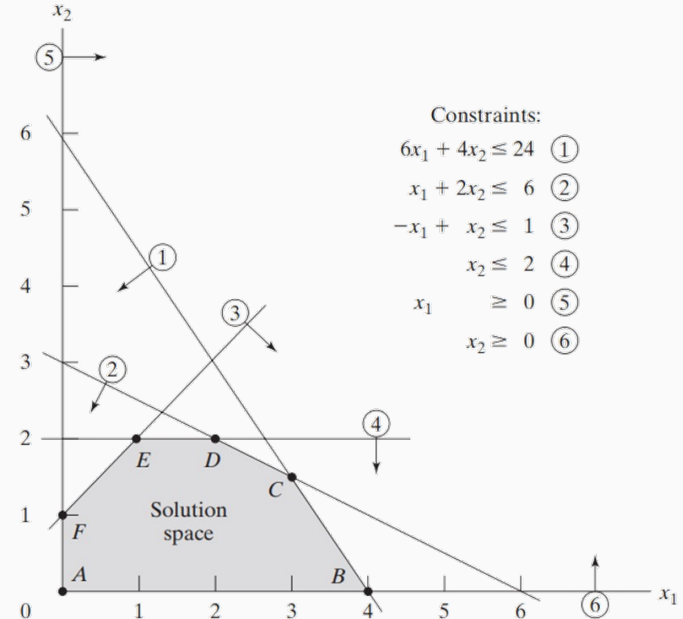
2 Linear Programming (LP) and graphical solution

3 Simplex algorithm for solving Linear Programming (LP)

4 Post Optimality analysis

5 Solving linear programming using solvers

6 Revision about linear programing



Course content (Part 2: Graph theory)

7

Introduction to graph theory and basic concepts

8

Paths and Connectivity in graphs

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Trees

10

Planar graphs

11

Partitioning and coloring problems

12

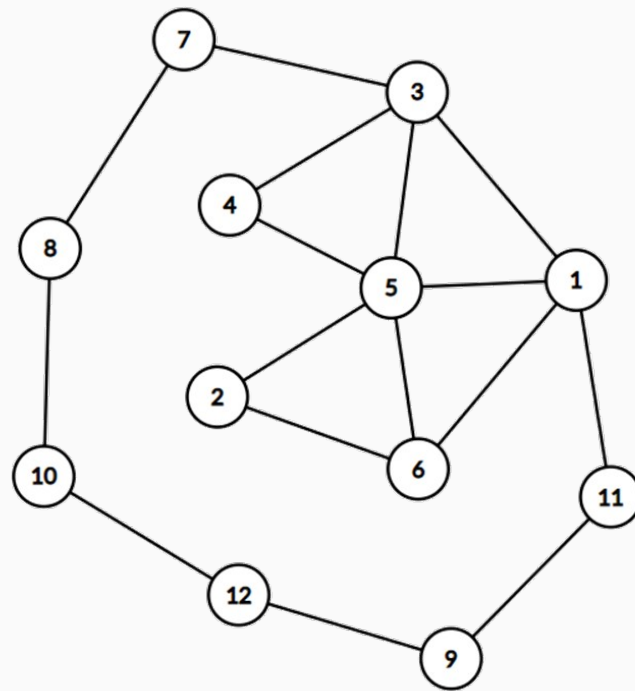
Shortest Path and Maximum Flow problems

13

Applications of Graph Theory to problems

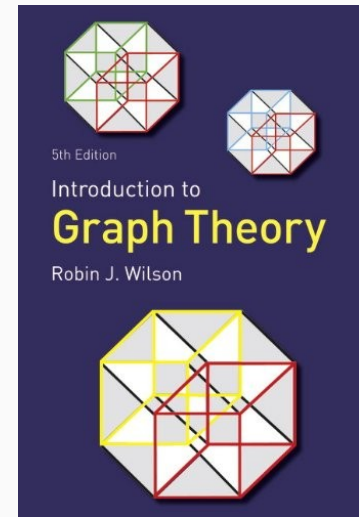
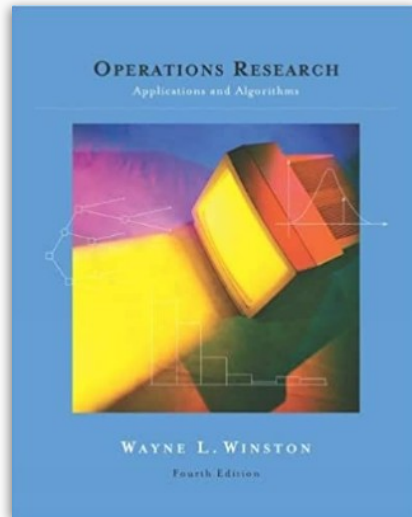
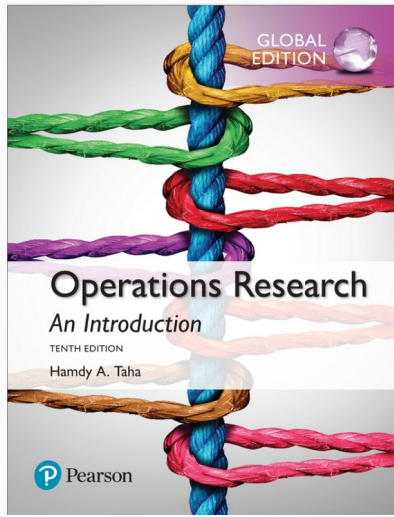
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Revision about Graph theory



Suggested text Books

The proposed textbooks are optional and only serve as a supplement to enhance your understanding of the course material.



Assessment method & Grading policy

- Exam - 60%
- Continuous Evaluation - 40%
 - Mid-Term Exam - 10 points
 - Quizzes/Homeworks - 3 points
 - Lecture Attendance - 3 points
 - Instructor Appreciation - 2 points
 - Tutorial Attendance - 2 points (maximum 2 absences)

Attendance & Participation Policy

- Regular attendance is expected
- Participation in class discussions and group activities mandatory
- More than 2 absences without justification may result in disciplinary actions
- Unexcused lateness to class will not be accepted.

Academic Honesty Policy

- All work must be original and completed to the best of ability
- **Plagiarism** and **cheating** will not be tolerated
- Appropriate disciplinary action will be taken for violations

***Education is not the filling of a pail, but the
lighting of a fire.***

التعليم هو إيقاد شعلة، وليس ملء وعاء - سقراط